Acoustics Lab Four Report

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Abstract:

This lab introduces an experiment and theory regarding a Helmholtz resonator. This will be used to determine the resonance frequencies at different volumes and effective length of the neck compared to the physical neck.

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I. Introduction:

The objective of this lab is to determine the resonance frequencies of a Helmholtz resonator experimentally as the volume is changed. The resulting data collected will be used to determine the effective length of the neck and compare it to the physical hand measurement of the neck.

Equipment List & Setup:

ITEM	NEEDED
Microphone	1
Ring stand and clamp	1
Digital oscilloscope	1
Digital calipers	1
100 mL beaker	1
2 L boiling flask	1

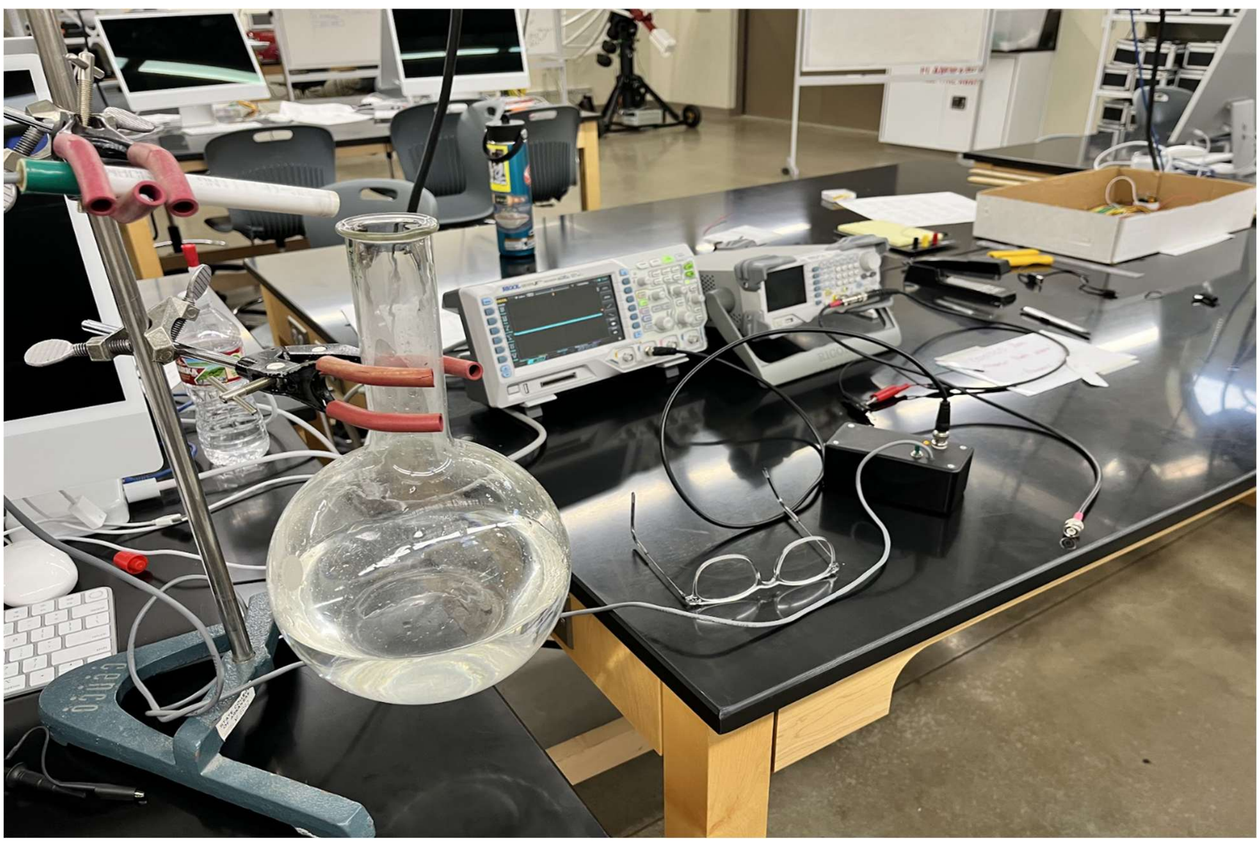


Figure 1: Experimental Setup and Helmholtz resonator.

II. Theory:

A. Background:

A Helmholtz resonator consists of a volume and a connecting duct (neck) as illustrated in Figure 2. We can use a lumped element approach to modeling the Helmholtz resonator if its physical dimensions are small with respect to the wavelength. In this case, the wavelength is the lowest frequency mode of oscillation, i.e. the Helmholtz mode. Recall from our discussion in class as well as reading the textbook that the Helmholtz angular resonance frequency is given by,

$$\omega_0 = c\sqrt{\frac{A}{V\Delta x}}$$

where A is the cross-sectional area of the neck, V is the volume of the cavity, Δx is the neck length and c is the speed of sound in air at room temperature (346 m/s). We can write the angular frequency as $\omega = 2\pi f$.

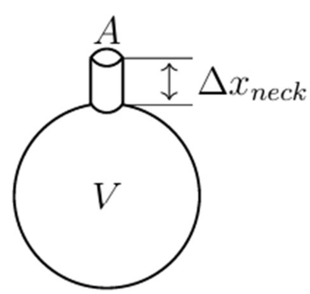


Figure 2: Physical dimensions of the Helmholtz resonator.

For ease of use we will adjust equation 1 as:

$$\Delta x_{eff} = \left(\frac{c}{2\pi f_{exp}}\right)\frac{A}{V}$$

III. Experimental Procedure:

We will be using another equation that looks like slope intercept form to do our calculations. We can experimentally measure f_{exp} for different volumes, V, and plot the data as the author of "Understanding Acoustics" does on page 380 Figure 8.17. The slope of the data is the effective neck length. Note: the experimentally measured frequencies are related to the period by T = 1/f, thus, we have:

$$\left(\frac{c^2}{4\pi^2}A\right)T_i^2 = \Delta x_{eff}(V - \Delta V_i)$$

where ΔV_i is the change in volume and T_i is the period for the i-th measurement. This theoretical relationship looks like the equation for a straight line: $y = mx + b$.

The 4 steps to this lab are below:

1. Take approximately 10 data points by gently injecting water into the flask volume and measuring the resonance frequency by blowing over the top of the flask.

2. Use your data to make a meaningful plot so that you can make use of Eq 3 to do the analysis.

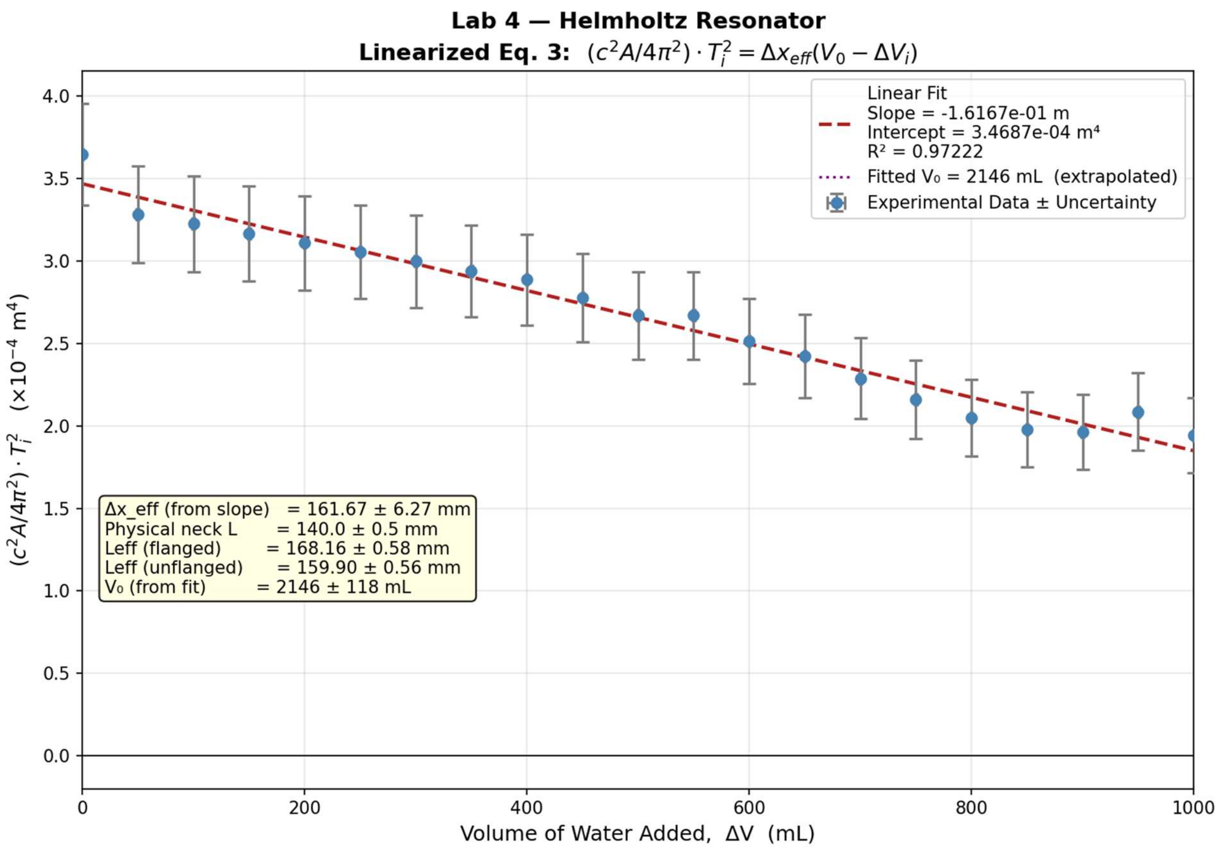
3. Use an appropriate method to determine the slope and intercept with uncertainty of the data. Determine Δx_{eff} from the fit.

4. Measure the physical length of the neck with uncertainty. Compare the physical neck to the effective neck length. Are they the same within uncertainty bounds? Why or why not?

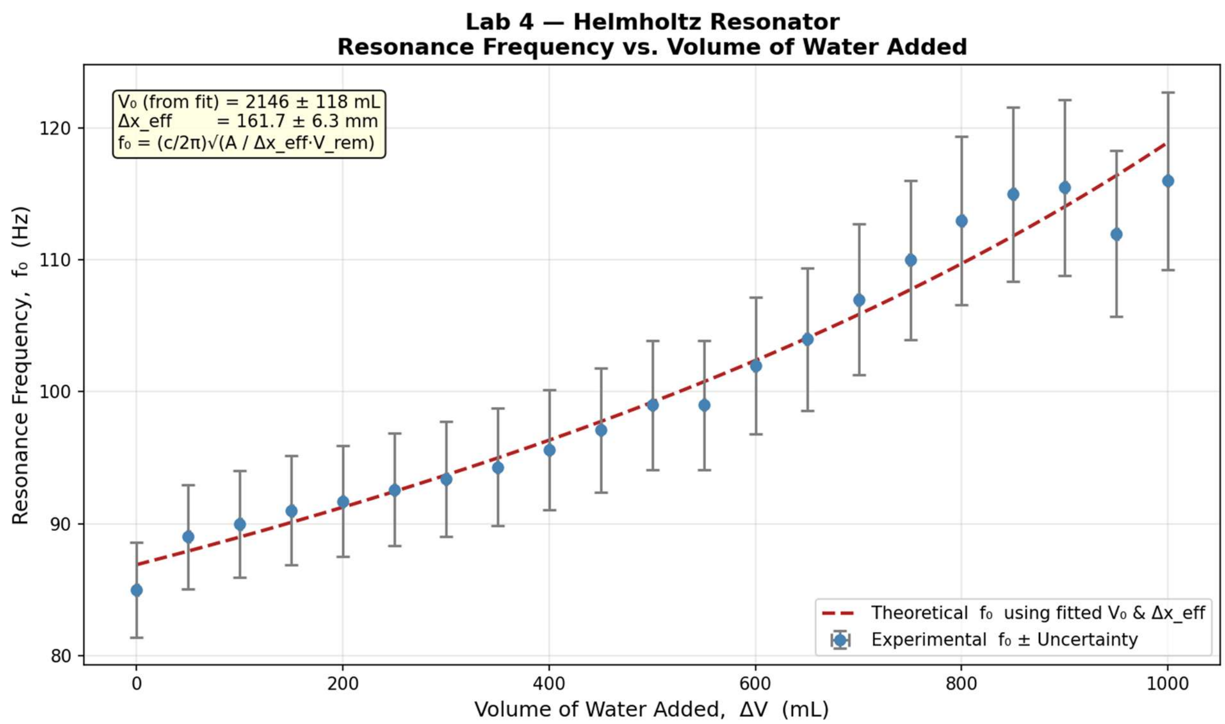
IV. Results:

Results from this were as expected. Using CoPilot in VSCode gave me a confusing plot of data at first. After some work I got the results below. First the linearized equation plot then the second plot of resonance frequency vs. volume added which is much easier to understand.

A. Linearized Equation:



B. Resonance Frequency vs. Volume added Plot:



C. Resulting Data:

Physical Parameters

Speed of sound (c): m/s	345.9 m/s
Neck diameter: mm	33.17 mm
Neck radius (a): mm	16.59 mm
Measured Neck length (L): mm	140 mm
Cross sectional area (A): m^2	$8.64 * 10^{-4}\ m^2$

Linear Fit Results

Slope: m	$-0.162 \pm 0.006\, m$
Intercept: m^4	$3.469 * 10^{-4} m^4$
R^2:	0.972
Δx_eff (slope): mm	$161.7 \pm 6.3\, mm$
Δx_eff (intcpt): mm	$161.7\, mm$ uses fitted V0
Physical neck L: mm	$140.0 \pm 0.5\, mm$

Length Corrections

Physical L:	$140.0 \pm 0.5\, mm$
L_{eff} (flanged):	$168.2 \pm 0.6\, mm$
L_{eff} (unflanged):	$159.9 \pm 0.6\, mm$
Δx_{eff} (from fit)	$161.7 \pm 6.3\, mm$

Length Corrections

Δx_{eff} vs. Physical L:	*|161.67 - 140.00| = 21.67 mm vs combined unc = 6.77 mm → ✗ DISAGREE*
Δx_{eff} vs. L_{eff} (flanged):	|161.67 - 168.16| = 6.48 mm vs combined unc = 6.85 mm → ☑ AGREE
Δx_{eff} vs. L_{eff} (unflanged):	|161.67 - 159.90| = 1.77 mm vs combined unc = 6.83 mm → ☑ AGREE
Δx_{eff} (from fit)	$161.7 \pm 6.3\, mm$

Flask Volume from X-Intercept

V_0 from Fit:	$2145.5 \pm 117.7\, mL$
Estimated Volume:	~2000 mL (2L flask)
Difference from estimate:	146 mL (7.3%)

V. Discussion:

A. Overall:

The resulting plots after corrections looked really good. It can be confusing to understand the y-axis on the first plot. All that is, is the indirect frequency based on the period. Think of it as the mathematical trick to get a straight line. In doing this we can extract the slope and intercept.

Performing the experiment was fun especially cracking jokes while doing it. However, I must note that the data collection was a bit tedious. Figuring out the right angle, position, and amount of air to get the flask to resonate was difficult. One of my classmates played clarinet and was the best at getting the flask to produce a consistent sound. It makes you think about how instruments work in a similar fashion.

Relating to the statement of tedious data collection, another thing we delt with is the measured frequency on the oscilloscope not being very consistent. Blowing air into the flask produces waves on the oscilloscope that vary in amplitude, frequency, etc. Some extra noise was likely captured while collecting data which could affect results. Another thing to note is while we had consistently rising resonant frequencies, the measurements started to backtrack once we got to about 850 mL of water added to the flask. For example, at 900 mL we had a frequency of about 115.5 Hz, but at 950 mL we had a frequency of 112 Hz. Then at 1L we had a frequency of 116 Hz. These are weird quirks caused by human error in data collection.

Another human error that I noticed was the measurement of the length of the neck. It was recorded as 140mm with an error of +/- 0.5mm. The measurement was taken quickly, outside the flask and did not consider measuring from the inside of the neck. Measuring from the inside would allow us to measure from the top all the way to the part where the neck starts to curve inward with the rest of the round flask.

Overall I am happy with the results that were produced. If I were to do this lab again I'd like to set a trigger properly on the oscilloscope or collect the raw data straight from it. This way we are not watching oscillating frequencies and guessing.

B. AI-Usage:

The usage of CoPilot inside of VSCode is great. Parts of this lab in my opinion need to be done with your own personal knowledge and reasoning skills. It is important to do this during experimentation as understanding where human error occurs or could happen plays a huge role in experimentation in all areas of study. Now once you've collected good sufficient data and know where faults might lie, I believe it is totally reasonable to use AI in creating graphs and summarizing data. Make sure to review its results as I had to correct a few things in mine.

VI. Conclusions:

I have no grand conclusions about the lab. The data collected agreed with the theoretical estimations. It also showed how the length of the neck is not simply a measurement you make with a ruler. The column of air that comes out of the neck stays in the shape of a cylinder for a little bit then as air resistance takes over it spreads out in a cone shape. I imagine the sound produced by performing this lab is similar to a water bottle that's full, and when you turn it upside down the water pours out. However, air must return back into the bottle to fill that missing volume. It oscillates until the bottle is empty and the volume is filled with air. The air traveling into the tube implies that air must travel out. When the resonance frequency is met, it starts to oscillate like the water bottle being turned upside down.

VII. References:

Note: I need help with proper referencing

[1] "Lab 4: Helmholtz Resonator", Dr. Slaton Ph.D., 2026

[2] CoPilot in VSCode, Creating plots and calculations.

[3] Understanding Acoustics - An Experimentalist's View of Sound and Vibration, 2nd Edi tion, by S. Garrett, Pg. 380, Figure 8.17